

ABDULLAH MANZOOR MALLICK

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PROFESSIONAL SUMMARY

Final-year Computer Science student with hands-on experience in software engineering, AI/ML, big data pipelines, blockchain/Web3, compiler design, and systems programming. Completed a Java backend internship at AKSA Solutions Development Services building full-stack Spring Boot systems, alongside delivering production-quality independent projects: a Zero-Knowledge Proof smart wallet on Ethereum (FYP), a real-time Lambda Architecture crime analytics system using Apache Spark and Kafka, and an autonomous TORCS racing AI trained with scikit-learn neural networks. Strong foundation in data structures, algorithms, operating systems, and OOP. Available for internships and full-time software engineering, AI/ML, or data engineering roles.

TECHNICAL SKILLS

Languages: C · C++ · C# · Java · Python · JavaScript · x86 Assembly · SQL · Solidity

Backend & AI/ML: Spring Boot · Spring Security (custom JWT/RBAC) · Spring WebFlux/R2DBC · scikit-learn (MLP · RF · DT · KMeans) · PyMC · NumPy · Pandas · PySpark MLlib

Big Data & Messaging: Hadoop (MapReduce · HDFS) · Spark (PySpark · Spark SQL · MLlib) · Kafka · Spring Cloud Stream · Storm · Docker

Blockchain / Web3: Solidity · Hardhat · Foundry · Circom · Groth16 · Ethers.js · Viem · ERC-20 · Ethereum Sepolia · Polygon Amoy

Web & UI: HTML5 · CSS3 · JavaScript · React · Thymeleaf · Java Swing + JDBC · Streamlit

Databases: MySQL · PostgreSQL · MongoDB · SQL Server · SQLite · R2DBC

DevOps / Cloud: Jenkins CI · Microsoft Azure (App deployment) · Docker · Docker Compose · Git/GitHub

Systems & Compilers: OS concepts · x86 Assembly · BSD Sockets (C) · Lexical analysis (DFA/NFA) · LL(1) · LR(1)/SLR(1) · Flex & Yacc/Bison

Security: AES · DES · RSA · PKI · SQL injection · XSS · CSRF · Penetration testing · CTF

EXPERIENCE

Summer Intern (Java) — AKSA Solutions Development Services (Pvt.) Ltd. *Islamabad, Pakistan | June – August 2026*

- Independently built 8 full-stack Java/Spring Boot backend projects to demonstrate applied understanding of production workflows — including custom JWT authentication with fine-grained RBAC (no Spring Security), reactive programming (WebFlux/R2DBC), concurrency control via pessimistic and optimistic locking, and event-driven architecture with Kafka — each reviewed and validated by a senior engineer.
- Presented and explained each project's design decisions and trade-offs directly to a senior developer, translating self-taught concepts (SOLID principles, layered architecture, JWT security flows) into working, reviewed code.
- Worked solo across the 6-week internship, owning the full build-and-explain cycle for each project.

PROJECTS

Sentinel Z – ZK-Gated Account-Abstraction Wallet [FYP] *Solidity · Circom · Groth16 · React · Hardhat · Ethers.js | Spring 2026*

- Engineered a tiered-authorization Ethereum smart wallet: ZK-Proof verification (Circom + Groth16) for high-value transactions; standard key-sig for low-risk spending with daily-spend caps enforced on-chain.
- Built React wallet UI/UX, end-to-end test suite, and documentation; full ERC-20 and dApp (Uniswap) compatibility; deployed on Ethereum Sepolia and Polygon Amoy testnets. Team of 3.

Real-Time Crime Analytics & Alert System [Big Data Analytics] *PySpark · Kafka · Storm · PostgreSQL · MongoDB · Streamlit · Docker | Spring 2026*

- Lambda Architecture on Chicago Public Safety dataset (5.7 GB, 7M+ records): PySpark batch layer with K-Means clustering of 7M+ GPS coordinates; Kafka + Storm speed layer with sliding-window anomaly detection.
- Serving layer: PostgreSQL + MongoDB + Streamlit dashboard; fully containerised with Docker Compose; sub-second streaming latency alongside millisecond historical queries.

Helix — Distributed Banking Microservices *Java · Spring Boot · Jenkins · Azure · Docker*

- Designed a microservices-based banking system and configured a Jenkins CI/CD pipeline for automated build and deployment.
- Deployed to Microsoft Azure to validate the end-to-end pipeline in a live cloud environment (subsequently taken down after free-tier credit expiry).

Inventra — Inventory & Warehouse Management System *Java · Spring Boot · MySQL · Thymeleaf · Custom JWT*

- Built inventory/warehouse management with a manually implemented JWT authentication layer (deliberately built without spring-boot-starter-security) and fine-grained, per-admin activity-based RBAC rather than blanket role access.
- Modeled products, suppliers, warehouse stock, and incoming/outgoing order workflows with a role-aware server-rendered dashboard.

CryptoTracker — Reactive Live Price Streaming Service *Spring Boot · WebFlux · R2DBC · JPA · MySQL · JWT*

- Built a fully non-blocking reactive pipeline: WebClient polls CoinGecko's live API every 30s, writes via R2DBC, and streams updates to clients over Server-Sent Events.
- Combined R2DBC (high-frequency reactive writes) with JPA (low-frequency batch archival) in the same service, isolating blocking work on a dedicated scheduler thread to avoid contending with the reactive event loop.

Weather Aggregator *Spring Boot · PostgreSQL · WebSocket · JWT*

- Aggregated live weather data from the Open-Meteo API, comparing three HTTP client approaches (URLConnection, RestTemplate, WebClient) and pushing real-time updates to subscribed clients via WebSocket (STOMP/SockJS).

Library Management System *Java · Spring Boot · MySQL · JWT · Docker*

- Full-stack library system applying SOLID principles and 6 classic design patterns (Factory, Strategy, Builder, Observer, Repository, Singleton), with rotating JWT refresh tokens, WebSocket + email notifications, and a 22-test automated suite.

Custom Language Compiler [Compiler Construction] *Java · Flex · Yacc/Bison · C | Spring 2026*

- Hand-coded lexical analyzer (RE → NFA → DFA) with JFlex validation; full LL(1) predictive parser and SLR(1)/LR(1) bottom-up parsers with canonical-item construction; JSON→XML translator via Flex + Yacc/Bison with full AST traversal.

Autonomous Racing AI – TORCS Simulator [Artificial Intelligence] *Python · scikit-learn · NumPy · Pandas · Pygame | Fall 2025*

- Sensor-driven AI racing agent trained via deep MLPRegressor on 19 track sensors across 14+ car-track datasets; full ML pipeline from data collection through StandardScaler, train/test split, MSE/R² evaluation, and adaptive gear-shift logic. Team of 3.

Bayesian Statistical Modeling [Statistical Modeling] *Python · PyMC · ArviZ · Pandas · Matplotlib | Spring 2026*

- Bayesian inference (Beta-Binomial), linear regression with posterior predictive checks, and 5-model Bayesian logistic regression for student classification with decision-boundary comparison.

EDUCATION

FAST University NUCES, Islamabad *Fall 2022 – Present*

BS Computer Science

Coursework: Data Structures & Algorithms · OOP · OS · Databases · Computer Networks · Compiler Construction · Software Engineering · AI · Big Data Analytics · Statistical Modeling · Information Security · Assembly Language · Numerical Computing · Cloud Computing

CERTIFICATIONS

- AWS Cloud Practitioner Essentials — Amazon Web Services
- Pre-Security Path — TryHackMe (THM-ASMDQCTJKT)
- C++ Programming — ICP Institute
- C++ Certification — Programming Hub
- OOP Concepts in C++ Certification — Great Learning
- Build Your Portfolio Website with HTML and CSS — Coursera

ADDITIONAL

- Leadership: Scrum Master & Product Owner (Software Engineering project); team lead across multiple group projects.
- Version Control: active GitHub (Abdullah-07s); Git branching, pull requests, collaborative workflows.
- Cloud & DevOps: AWS Cloud Practitioner Essentials; Jenkins CI/CD pipeline configuration; Azure deployment experience; containerised deployment via Docker Compose.
- Languages: English (professional fluency) · Urdu (native).